Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · UE25CV141A Unit 3 Class 55 AV Summary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE25CV141A Unit 3 Class 55 AV Summary

1 UE25CV141A Unit 3 Class 55 AV Summary

Study focus: UE25CV141A_Unit 3_Class 55_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000166 · EV0000167

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Civil Engineering from the Department of Science and Humanities, PS University.
- This is class number fifty-five.
- In the last class, that is class number fifty-four, we had discussed about how to determine the shear force and bending movement for a beam which is loaded with different types of loadings.
- We will continue.
- the same topic in this class also, and we will see how to solve some more problems which are loaded with different types of loadings.
- And we are dealing with the topic under unit three, that is distributed forces, and in that chapter five, and the topic is beams internal effects.
- And we have referred to the textbook, Engineer Mechanics, Statics, 8th edition, the authors are Miriam Craig and Bolton.
- For today's class I have chosen this problem that is problem number 5 bar 127.
- The statement of the problem goes like this.
- Draw the shear and bending moment diagrams for the beam subjected to the end couple.
- What is the bending moment M at a section 0.5 meter to the right of beam.
- which is subjected to a couple at the end of the beam, that is to the right side of the beam.
- And also we need to find out what would be the value of the pending moment.
- At a section 0.5 meters from the right side of the support B.
- Which is supported at point A by a pin or a hinge support and at B by a roller support.
- The distance between A and B is 2 meters.

- At the end of the right side of the beam,.
- There is a couple or a moment that is acting in the clockwise direction of magnitude 120 newton meters.
- As well as we need to draw the bending movement diagram for the beam.
- After drawing the shear force and bending moment diagram, we are supposed to find out what is the bending moment acting at a point right side of B at a distance of 0.5 meters from B.
- That is this distance, if you take a section here.
- This distance has to be 0.5 meters and we need to find out what will be the bending moment at that point.
- And it is the beam is supported at A by mean support under log support.
- First thing that we need to do to solve this problem is to determine the external support reactions acting at A and acting at B.
- We will first draw the free body diagram by replacing the supports by their respective reactions.
- Then there is an horizontal reaction acting at A, we will assume it to be acting to the right.
- Then at B point we have a roller support.
- Reaction at B is upwards only.
- Whereas the sense or the reaction at A can be upwards or downwards, as well as it can be to the right or to the left because it's an hint support.
- When I take the algebraic sum of all the forces acting along the x axis that is horizontal and equality to zero, then I will get the reaction at A along the horizontal which should be equal to zero because there is no horizontal force that is acting.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then substituting the values between this.

Key details

- section that is between A and B and B and C.
- We can get the different values for shear force and body moment at any particular point between A and B and B and end of the V.
- That's what is done here.
- We have I have written the free body diagram here.
- The hint support is replaced by its reaction, vertical reaction acting upwards.
- I have assumed it to be VA and horizontal reaction acting at A, I have assumed it to be HA.
- Then the reaction at B which is the because of the roller support and it will be acting upwards.
- Then there is a movement acting at the free end of the beam, which is 120 newton meters, and that is what is shown.
- Equilibrium that is used R sigma M at A equals 0, as I have explained.

- Using that I have determined what should be the reaction that is going to be acting at B, which we will get it as 60 kilonewton.
- Then using the algebraic sum of all the forces acting.
- along the vertical, we will find out what will be the reaction at A and we will find that the reaction at A is acting downwards BA.
- Then we will also find out what is the horizontal reaction at A by taking all the forces along X, but we found it to be 0.
- Next, we will take a section in between A and B, okay, and that distance from A to be equal to X, that is what we will take.
- We will take all the forces acting to the left side of the section and we will determine the shear force equation.
- All the forces acting to the left, what we have is only the reaction acting at A, that is VA.
- Which is equal to minus R B, which is equal to 60 kilonot minus R B and it is acting downwards.
- To the left and downwards, you will get it as negative minus 60, and that's what is indicated here.
- And that value will remain constant for the entire section between A and B.
- Next, if you take a section B C here, okay, any section the distance from B is X, if you take what will be the shear force.
- The values are same, so we will get the shear force to be equal to zero.
- Using these two equations we can draw the shear force diagram as indicated here.
- Between A and B, it's a constant value and it is minus 60, so that is what is indicated.
- This is the datum line 00.
- Then we have a 60 between this, which is a constant value, and then between the section B and the end of the beam.
- it is the shear force is zero, so we have we have drawn that shear force diagram.
- We will take a section here.
- This distance is X.
- Then we will write what will be the bending moment.
- VA I have assumed it to be acting upwards.
- plus 60 into x, so we will get it as negative minus 60 into x because that will produce a hawking bending moment.
- Similarly, if I take a section between B and C, whereas this is C and this is B, this is X distance, and we will take all the moments acting or all the forces acting to the left of the beam and taking movement of all these forces, we will get bending moment to be equal to minus 120, that is because.
- we will have this 60 into this distance that is 2 plus x, whereas this 60 into x we will get.
- Or if I take the section to the right side of the beam.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 10:00–14:59

We will get a movement which is equal to 120, which will produce a hawking bending movement, so it is minus 120.

Key details

- But the support A and the free end it is 00.
- As we progress from A to B, as X increases, your bending moment also increases by 60 times X.
- And between B and C that is N of the beam, it's a constant value which is one twenty and it is negative, so it will get a constant value.
- So drawing this, we will get this diagram and this is negative because it is below the datum line and this is the bending moment diagram.
- The next part of the problem is to determine what would be the bending moment value at a distance 0.5 meters from B to the right side, that is at this, this distance is 0.5.
- that the bending moment is a constant value which is 120 that is negative bending moment.
- However, however we can get that.
- Then the shear force diagram and bending moment diagram together.
- We have determined the support reactions, which we have got it as 60 and 60 here, but at E it is acting downwards, at B it is acting upwards.
- Bending moment at point five meters, we will get it to be one twenty, that is what is indicated here.
- I hope all of you have understood this.
- We will move on and to the next.
- Question I have considered a question 59.
- Draw the shear and bending or draw the shear and movement diagrams for the loaded breed.
- What are the values of the shear force and bending moment at the middle of the beam is what is ask.
- And after that, we have to determine what would be the value of the shear force and bending moment at the middle of the beam.
- It's a simply supported beam.
- At the left hand, that is at A, we have an inch support or a pin support.
- At the right hand, B we have a roller support.
- Then we have a point load acting downwards of magnitude 4 kiloneeton at a distance of 1 meter from A, which we will take it as point C.
- Then we also have a counterclockwise moment of magnitude 2800 newton meters acting at a this point, which we will take it as D, which is at a distance of 2 meters from A, whereas 1 meter from C and as well as 1 meter from B to the left side.
- So, again in this problem, if I want to draw the shear force on bending moment diagram, first thing that we need to do is to draw or to draw the free body diagram and then to determine the support reaction's value.
- Hinch so two reaction, one vertical and one horizontal, but B there is roller, always it is going to be acting upwards because the roller as you can see it is indicated like this.
- I have replaced the.
- Heat support by two reactions, vertical and horizontal, then roller support at B by one reaction RB and I have indicated the load that is acting on the beam.

- Then first equations of equivalent that I have taken is sigma fx is equal to 0, with which I will get movement or the horizontal force at A to be equal to 0.
- Then the next equation taken as algebraic sum of all the vertical forces to be equal to 0.
- 4 kilonut which I have converted into newtons which is 4000 acting downwards, so negative.
- Then we have R B acting upwards or positive, R B equals 0.
- So what we will get is we will get 4000, that is because of this point load into 1 meter.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 15:00–19:59

Which is clockwise, so negative.

Key details

- Then we have this movement which is anti-clockwise, so 2800 is acting in anticlockwise direction, so positive.
- Then R B will also produce a counterclockwise movement, so that is also positive.
- From this equation we will get the value of R B which is equal to 400 newtons.
- We will substitute this 400 newtons in this equation and get the value of VA, which will be equal to 3600 newtons.
- Reaction at A vertical reaction is 3600 newtons.
- Vertical reaction at B is 400 newtons.
- And both the reactions at A and B which are vertical are acting upwards itself.
- So shear force is equal to Va because there is only one reaction that is acting to the left of the beam that is VA, so that's a constant value which we will get as 3600 as indicated here.
- Next, we will take a section between C and D, that is here.
- If you take the left side of the section, we will get shear force equals 400, that is VA, which is acting upwards, 3600, and 4000 newtons, which is acting downwards, so negative.
- between the section C and D.
- From A is X that we have taken.
- Minus 4000 into X minus 1.
- Because I have taken the X distance from A to this point as X, it is.
- X minus 1, which will be this distance.
- This is the datum line 00.
- Between this section, it's a constant value of 3600.
- Then coming to this section, again, shear force is a constant value which is minus 400.
- Then till the.
- C to B, it is constant itself because there is no vertical force acting between this.
- This is a shear force diagram.
- Above datum line is positive, below datum line is negative.
- The bending movement value is also increased, and if I put x is equal to 1, I get it as 3600, that is a point B, so it is a linear line because it's a point load.
- Then between C and D.

- Also we have this equation substituting the value of x, anything more than one and less than two, we will get the variation at x is equal to zero, we get it as 3600 and when x is equal to 2 we get the value of 3200.
- Then considering the moment we will get it to be 400 itself, and that is what we have indicated here.
- When we take the section B and D we get it as 400 when X is equal to 0 we will get it this to be 400 and that is what is indicated and when x is equal to 0 we will get x to be 0 when x is equal to 1 meter because we have taken it to the right side we will get this to be 4.
- Then he has asked you to determine what will be the value of the shear force embedding moment at the middle of the beam, which will be at a distance of 1.5 meters from A as well as 1.5 meters from B.
- From the shear force embeddement moment, if I take it.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 20:00–24:59

I the shear force value is four hundred, whereas the bending moment value we have to compute.

Key details

- It is varies from three thousand six hundred to three thousand two hundred.
- Or 3600 minus 3200, which is 400, and since it is the middle, it will be 200 mm, which is lesser than the value at B and 200 m 200 newton millimeters greater than the value at C.
- I'll explain this clearly.
- My name is Dr.
- Svintesh from Civil Engineering under the Department of Science and Humanities PS University.
- Discuss in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE25CV141A_Unit 3_Class 55_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 55_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 55_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 55_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.

4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.
9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00042	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00043	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00044	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00045	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

C Revision and retrieval practice